

FieldComms User Guide Supplement

NEW & EXPANDED FEATURES — v1.2

Sections 28 (expanded) + 31–35 · June 16, 2026

This supplement expands the original FieldComms User Guide (v1.0) with six sections: an expanded Kiwix Offline Library section covering content installation, the tier system, and adding new ZIM files; plus five sections covering features added after v1.0 was published. Replace Section 28 of the main guide with the expanded version here.

SECTION	TITLE	PAGE
28 ★	Kiwix Offline Library — Expanded (replaces main guide p.27)	2
31	Mode Switcher — Amateur Radio / ICS Modes	5
32	Reference Library	6
33	Operator Identity System	8
34	Offline Map Tile System	10
35	GPS Live Position	11

28. KIWIX OFFLINE LIBRARY — EXPANDED

<http://192.168.50.1:8081/>

Kiwix serves a complete offline reference library directly from the Pi on port 8081. Every device connected to EMCOMM-NET can open any Kiwix content in a browser — no internet required, no app to install. Content is stored as ZIM files (compressed, self-contained web archives). FieldComms includes a curated EmComm content catalogue organised into three tiers by disk space and operational value.

ACCESS KIWIX

Any browser on EMCOMM-NET → <http://192.168.50.1:8081>
Or from the FieldComms dashboard → click the **Kiwix Library** card

EmComm Content Catalogue

TIER	NAME	SIZE	WHAT'S IN IT	BEST FOR
1 Essential	WikiMed Medical Encyclopedia	~471 MB	Symptoms, diagnoses, treatments, medications, procedures, anatomy, lab values	Mass-casualty events, medical support, welfare checks
	Wikipedia (Mini) — Compact Reference	~1.2 GB	Compact English Wikipedia — facts, geography, history, science	General field reference, briefing preparation, situational awareness
	Wikivoyage — Shelter & Travel	~820 MB	Emergency shelters, evacuation routes, local resources by city and county	Shelter-in-place planning, unfamiliar jurisdictions, mutual aid areas
2 Recommended	Wikibooks — Field Manuals & How-To	~4.2 GB	First aid, survival, ham radio, electronics, cooking, field skills — all how-to	Training reference, field skills refresher, operator self-help
	iFixit — Equipment Repair Manuals	~3.1 GB	Step-by-step repair guides for radios, laptops, generators, medical equipment	Field equipment repair without manufacturer docs or internet
3 Enhanced	Wikipedia Medicine — Full Clinical	~8.5 GB	Complete medical Wikipedia: diseases, drugs, surgery, pharmacology, anatomy	Advanced medical support, extended operations, clinical reference
	Wikiversity — Training & Education	~3.4 GB	ICS courses, CERT training, ham radio education, first responder materials	Training sessions, exercises, certification prep, new operator onboarding
	Electronics Stack Exchange	~2.2 GB	Q&A; for radio circuits, antenna design, power systems, embedded electronics	Technical troubleshooting, antenna problems, radio repair

Tier 1 ~2.5 GB total ✓ Fits on any 8 GB Pi SD card

Tier 1 + 2 ~10 GB total ✓ Fits on 16 GB SD or USB drive

All Tiers ~22 GB total Requires 32 GB SD or USB drive

NOTE Storage is the main constraint. WikiMed (Tier 1) alone is ~471 MB and is the most critical content for EmComm. Always install WikiMed first. Tier 2 and 3 content is best stored on an external USB drive rather than the Pi SD card.

Installing ZIM Content

ZIM files are installed using the `kiwix_setup.sh` script included with FieldComms. The script downloads from the official Kiwix repository, registers each file with `kiwix-serve`, and restarts the service automatically. You must be connected to the internet to download — do this before deployment, not during an activation.

COMMAND	WHAT IT DOES
<code>sudo bash kiwix_setup.sh</code>	Interactive menu — choose individual ZIMs or a full tier
<code>sudo bash kiwix_setup.sh --tier 1</code>	Install all Tier 1 content (WikiMed + Wikipedia Mini + Wikivoyage, ~2.5 GB)
<code>sudo bash kiwix_setup.sh --tier 2</code>	Install Tier 1 + 2 content (~10 GB total) — recommended for most deployments
<code>sudo bash kiwix_setup.sh --tier 3</code>	Install all content — all 8 ZIMs (~22 GB) — requires large storage
<code>sudo bash kiwix_setup.sh --list</code>	Show all available ZIMs with size, tier, and install status
<code>sudo bash kiwix_setup.sh --status</code>	Show installed ZIMs, file sizes, service status, and last update
<code>sudo bash kiwix_setup.sh --update</code>	Check for newer ZIM versions and offer to re-download updated files

TIP Download Tier 1 content at minimum before every deployment season — WikiMed in particular is updated regularly with improved medical content. The `--update` flag checks for newer versions without re-downloading unchanged files.

Adding a Custom ZIM File

Any ZIM file from kiwix.org can be added to the library — not just the pre-catalogued ones. You might want to add a county-specific resource, a specialised medical reference, or a ham radio technical manual that becomes available after this guide was published.

- Download the ZIM file from **kiwix.org** or a Kiwix mirror to any device with internet
- Copy the file to the Pi: `scp mybook.zim fieldcomms@192.168.50.1:/opt/kiwix/zim/`
- Register it: `sudo kiwix-manage /opt/kiwix/library.xml add /opt/kiwix/zim/mybook.zim`
- Restart Kiwix: `sudo systemctl restart kiwix`
- Open **<http://192.168.50.1:8081>** — your new content appears in the library

File Locations & Service Management

ITEM	PATH / COMMAND
ZIM files	<code>/opt/kiwix/zim/</code>
Library index	<code>/opt/kiwix/library.xml</code>
Port	<code>8081</code> (access via <code>http://192.168.50.1:8081</code>)
Service status	<code>sudo systemctl status kiwix</code>
Restart service	<code>sudo systemctl restart kiwix</code>
View service log	<code>sudo journalctl -u kiwix -f</code>

Check disk usage

```
du -sh /opt/kiwix/zim/*
```

TIP Kiwix has a full-text search engine built in. Type any medical symptom, procedure, piece of equipment, or topic into the Kiwix search bar and it searches across all installed ZIM files simultaneously. Results appear instantly — completely offline. WikiMed is your fastest offline medical reference during an incident.

NOTE Kiwix is a separate application from FieldComms — it runs as its own system service (kiwix.service) on port 8081. If the library page does not load, check service status: `sudo systemctl status kiwix`. The FieldComms Health Monitor (port 5051) shows Kiwix service state as a green/red dot in the dashboard sidebar.

31. MODE SWITCHER — AMATEUR RADIO / ICS MODES

<http://192.168.50.1/>

The dashboard has a two-button mode switcher bar directly below the ESV header. It reorganises all tool cards into context-appropriate groups depending on whether you are running an amateur radio net activation or an ICS incident. The selected mode is saved on your device and remembered when you return.

MODE	WHAT IT SHOWS	WHEN TO USE
■ Amateur Radio	Operations (Net Control, APRS, Winlink, Callsign Lookup, Resource Board), Public Safety (Roster, DMS, Pre-Flight), Forms & Traffic (NTS, ICS-213, ICS-214), Reference (Propagation, Grid Square, Repeaters, Kiwix, Cheat Sheets). ICS Platform accessible via launch strip at bottom.	ARES/RACES activations, NTS nets, weekly nets, EmComm exercises
■ ICS / Incident Command	ICS Platform entry (prominent, full-width), Operations (ICS sections, T-cards, Resource Board, Tactical Map), Personnel & Coordination (Roster, Pre-Flight, DMS, Facilities), ICS Forms & Documentation, Communications tools.	Formally declared ICS incidents, multi-agency events, ICS paperwork required

A mode indicator pill in the top-right of the page header shows the current mode: ■ **Amateur Radio Mode** or ■ **ICS Mode**. Tap it to switch. The mode is saved in browser localStorage and remembered between sessions.

32. REFERENCE LIBRARY

<http://192.168.50.1/refs.html>

The Reference Library is a document management system for all field reference materials — radio manuals, SWIC interoperability plans, ICS forms packages, SOGs, agency plans, and training documents. Files are stored on the Pi and accessible from any device on EMCOMM-NET without internet.

TAB	CONTENTS
■ Amateur Radio	Radio manuals, frequency plans, ARRL publications, band plans, antenna references
■ ICS / Emergency Mgmt	SWIC plans, NIMS documents, ICS forms packages, agency SOGs, mutual aid plans
■ All Documents	Every uploaded document regardless of category

Uploading a Document

- Click **■ Upload Document** in the top-right — a panel slides in from the right.
- Drag-and-drop a file or click Browse. Accepts PDF, Word, Excel, PowerPoint, images, KML, ZIP, GPX — up to 200 MB.
- Fill in Title, Category, Source/Publisher, Applies To, Description, Tags, Revision, and Expiry date.
- Under **Show on tab(s)** — check Amateur Radio, ICS / EmMgmt, or both. A document checked for both appears on both tabs simultaneously (cross-referencing) without storing two copies.
- Click **■ Upload Document**. PDFs get a thumbnail automatically if poppler-utils is installed.

Ten ICS position job aid PDFs (IC, Safety Officer, PIO, Liaison, Ops SC, Planning SC, Logistics SC, Finance SC, Net Control Station, Field Operator) have been pre-generated and are ready to upload to the ICS tab. Left sidebar filters by category, tag, and cross-reference status.

33. OPERATOR IDENTITY SYSTEM

Every FieldComms page has an identity system that tracks who is operating at each device. There is no password — identity is stored in the browser and remembered between sessions. The system supports all four member types in McHenry County ESV:

TYPE	IDENTIFIERS	BADGE SHOWN AS
ESV Member + Ham	ESV Member ID (ESV-042) + Callsign (K9ESV) + Starcom Radio ID (1042)	■ K9ESV Jim
ESV Member, Non-Ham	ESV Member ID (ESV-042) + Starcom Radio ID (1042) — no callsign	■ ESV-042 Jennifer
Visitor / Mutual Aid, Ham	Callsign + Agency name — no ESV number assigned	■ W9XYZ Bob
Visitor / Mutual Aid, Non-Ham	Name + Agency name — callsign and Radio ID optional	■ VISITOR Red Cross

Using the Identity Picker

- On first use, the ■ **Identify Yourself** picker appears automatically after ~1 second.
- Select **ESV MEMBER** or **VISITOR / MUTUAL AID** at the top.
- For ESV members — choose from the roster dropdown (groups: ■ Amateur Operators / ■ ESV Members), or enter Member ID, Callsign, and Starcom Radio ID manually.
- For visitors — enter name, agency, and optionally callsign or radio ID.
- Click ✓ **CONFIRM**. Saved on this device — remembered next time.
- The top-right of every page shows your identity badge. Click it to change.

Operator Access Cards (Avery 5371)

Pre-printed business cards (3.5" × 2", 10 per sheet) can be generated from the roster. Each card shows WiFi network, password, URL, and the operator's callsign or member ID. Ham operators: callsign large in EOC Blue. Non-ham members: member ID large in purple. All cards show ESV Member ID and Starcom Radio ID.

```
python3 /opt/fieldcomms/scripts/gen_operator_cards.py --ssid EMCOMM-NET --password YOUR-PASSWORD
```

34. OFFLINE MAP TILE SYSTEM

Map tiles are downloaded in advance from USGS or Esri, stored as MBTiles SQLite files on the Pi, and served locally on port 8083. Both the Tactical APRS Map and Starcom Resource Map use offline tiles by default — no internet required during an activation. Default: **USGS Imagery+Topo Hybrid** — public domain satellite imagery with roads, contours, and place names overlaid.

- `sudo bash /opt/fieldcomms/scripts/download_tiles.sh` — interactive menu with 106 presets
- Presets include all Illinois counties, Wisconsin/Indiana/Iowa border counties, all 50 US states
- Search: `--search "dupage"` · Download: `--area county_mchenry`
- Custom area: `--north 42.70 --south 41.90 --west -89.00 --east -87.80 --zoom 8-16`
- Downloads are resumable — interrupted downloads continue from where they stopped
- McHenry County recommended: Essential Z8–14 (~8 MB, ~2 min) · Standard Z8–16 (~60 MB, ~10 min)

35. GPS LIVE POSITION

USB GPS hardware integrates with FieldComms to provide live station position. When a fix is acquired, the Tactical APRS Map, NWS weather alert zone, and station coordinates all update automatically in real time.

FEATURE	WITH GPS FIX	WITHOUT GPS
Tactical APRS Map	Station marker moves to live GPS coordinates every 30 seconds	Marker shown at configured coordinates
NWS Weather Alerts	Alert queries use live GPS — shows alerts for your actual location	Uses configured county coordinates
Dashboard Sidebar	Shows live lat/lon, fix quality (2D/3D), and last fix time	Shows "No Fix" with configured coordinates
Supported Hardware	USB GPS receivers with CP2102, CH340, u-blox NEO-6/7/8, Adafruit, PL2303, FTDI chipsets — auto-detected by udev rules, appears as /dev/gps0	Coordinates set at install time or via API

NOTE GPS hardware is optional. Without it, FieldComms uses the coordinates configured at install time (McHenry County default: 42.3247, -88.3822 if not changed). All features work correctly without GPS — you simply get a fixed position rather than a live one.